Per-task gate threshold fitting

Qwen2.5-Omni-7B on OmniPro Online — what it is, what it found, and what it does not show

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1 The question, in one page

The system is a **proactive streaming assistant**. A video plays; the model watches and listens as it arrives; at some moment it should *speak* — report a count, name a state change, alert on an event. The research question is not *what* to say. It is **when**.

The machinery for deciding is small. Once per tick the system asks the frozen model a yes/no question — "do we have enough information / has the event happened?" — and reads the probability of `true` straight off the logits. Call it p_{hit}. If p_{hit} crosses a threshold, the system speaks.

That threshold is the object of this study. There are nine tasks, and the guess motivating the work was that they want *different* thresholds: a task that should alert once per video is not a task that should narrate continuously.

So the study asks: does fitting one threshold *per task* produce a better streaming system than a single threshold shared by all nine? And the answer turns out to be more interesting than a yes or a no.

The benchmark is OmniPro Online: 9 tasks × 300 videos = **2,700** samples, 24.1 s to 594.6 s long, 65% of them requiring audio. **The metric** is `time_f1`: an emission counts if it lands within ±3 s of a ground-truth trigger, matched greedily one-to-one. It needs no LLM judge, which is why every decision in this study is made on it.

2 Why it had to be done at all: the threshold was doing nothing

Before any fitting, an audit of the shipped system found the knob was **disconnected**.

Under the shipped `gate_strategy="hysteresis"`, the test that actually decides whether to fire was `p_hit >= gate_high_thr` — a fixed global **0.5**. The per-task `hit_threshold` only decided whether an answer got *decoded*. Across a completed 2,700-sample run (581,403 ticks, 50,433 fires) **not one fire occurred below** $p_{hit} = 0.500$. The four tasks configured at or under 0.5 were all silently running at 0.5. Two further parameters, `task_gate_modes` and `task_refractory_s`, were read **nowhere at all**.

This is the pattern the whole project keeps meeting: **the failure was silent**. It produced a finished run and a number you could put in a table. Fixing the gate so the threshold is genuinely in the firing path is what made a fit meaningful — and it means the "shipped" thresholds were never really tested at their stated values.

A second defect surfaced while the fit was already running. p_{hit} is *high on the very first tick* of every video — the model answers the question from the prompt before it has seen anything (≥0.05 for 98–100% of samples, median 0.34–0.64). Seeding the gate's `prev_above` to `False` made that first

tick a rising edge, so a task with a long refractory spent its single allowed emission at $t \approx 1$ s against ground truth at 26–98 s. The fix is definitional — an edge requires a predecessor, so `prev_above` starts `None`. Offline best time-F1 rose $0.0535 \rightarrow 0.0900$ (`snapshot_counting`) and $0.0636 \rightarrow 0.0751$ (`instant_event_alert`); short-refractory tasks moved by ± 0.001.
The size of that effect tracks the refractory — which is the practical reason the gate's three knobs (`threshold`, `mode`, `refractory`) cannot honestly be fitted one at a time.

3 The approach

Three stages, deliberately cheap before expensive.

stage	what it does	cells	sample-evals	status
pass 1	wide grid, 10 thresholds × 9 tasks	94	1,410	complete
pass 2	refine 5 points around each pass-1 winner	45	675	complete
stage 3	full benchmark at the fitted values	1	2,700	94%

The frozen subset. Passes 1 and 2 run on a fixed 5% all-audio subset — **15 videos per task, 135 in total** — chosen once and never changed. Freezing it is what makes the two passes comparable to each other and to the bootstrap that audits them.
Protocol rules that shaped every number here:

- **The bootstrap resamples *videos*, never ticks or events.** Ticks within one video are serially correlated through a shared KV cache; resampling them would manufacture confidence that is not there.
- **Paired.** Every threshold is scored on the *same* resampled video set, and the selection rule is re-run inside each draw rather than an argmax being taken — so the audit measures the stability of *the procedure*, not of a number.
- **Re-selection is always divided by chance.** A grid of 10 cells has a 10% chance rate, one of 5 cells has 20%. Comparing raw re-selection rates across grids of different size is meaningless.
- **A 0.03 noise band.** The eval is not bit-reproducible; run-to-run variance on one fixed config has been measured at F1 0.255 vs 0.051. Differences below 0.03 are not treated as real.
- **Content metrics are withheld, never estimated**, whenever the LLM judge is unreachable. A blank cell is a fact; a guessed cell is a lie that survives into a paper.

4 What the fit produced

task	shipped	**fitted**	gap vs rival	grid span	re-sel	chance
cumulative_counting	0.925	**0.6167**	−0.0017	0.048	48%	20%
dedup_counting	0.992	**0.5500**	+0.0243	0.055	70%	20%
event_narration	0.100	**0.6500**	+0.0037	0.023	74%	20%
explicit_target_grounding	0.500	**0.7167**	+0.0039	0.065	47%	20%
instant_event_alert	0.450	**0.4500**	+0.0645	0.065	63%	20%
realtime_state_monitor	0.800	**0.4500**	−0.0093	0.029	38%	20%
semantic_condition_alert	0.980	**0.2167**	+0.0091	0.066	42%	20%
sequential_step_instruction	0.010	**0.1500**	+0.0025	0.022	41%	20%
snapshot_counting	0.985	**0.3833**	+0.0667	0.133	56%	20%

The fitted values move a long way from the shipped ones — but read the next section before treating any of them as a discovery. `instant_event_alert` did not move: it scored **0.000 at every threshold** in pass 1, so the selection rule correctly flagged it FLAT and kept the shipped value rather than picking a winner out of noise.

5 The findings, in the order they arrived

5.1 Finding 1 — p_{hit} does not rank event-adjacent ticks above quiet ones

This is the central result. On the five tasks with a complete reference cell, the ROC of per-tick p_{hit} against a ±3 s positive label:

task	ticks	distinct p_{hit}	**AUC**
semantic_condition_alert	3,323	253	**0.431**
cumulative_counting	2,833	284	**0.469**
explicit_target_grounding	2,947	232	**0.512**
snapshot_counting	3,703	280	**0.526**
instant_event_alert	4,047	247	**0.551**

0.5 is chance. Every 95% CI contains it. And the obvious escapes were all checked and closed: it is not a clock offset (swept ±10 s — no peak), not too tight a tolerance (±10 s — unchanged), not a degenerate score (232–284 distinct values spanning 0.000–0.997).

Why this governs everything else. A threshold is a *ranking* device: it works only if the score is higher near events than away from them. If AUC is 0.5, the threshold controls *how often* the system speaks, never *when*. Every result below is this fact arriving again by a different route.

5.2 Finding 2 — no task's fitted threshold survives its own bootstrap

Pass 1 completed (1,410/1,410, 94/94 cells) and the rule returned a winner for all nine tasks. Then the paired bootstrap:

- **9 of 9** tasks have a ΔF1-against-best-rival CI **containing zero**.
- Re-selection is 25–61% against an 8–10% chance rate — `semantic_condition_alert` at just **6%**, i.e. *below* chance.
- Two "fits" are artefacts: `instant_event_alert` scores 0.000 everywhere, and `explicit_target_grounding`'s entire curve is one matched event or zero out of 16.

The only reproducible structure in the whole grid is the collapse at 0.95 — i.e. the threshold controls emission *volume*, exactly as an AUC of 0.5 predicts.

5.3 Finding 3 — the gate has one identifiable degree of freedom, not nine

A fair arena, all arms scored on the same videos at equal budget:

arm	time-P	time-R	**time-F1**	vs. best single global
`shipped_per_task`	0.151	0.356	0.2115	−0.0354, CI [−0.055, −0.016]
`global_0.5`	0.142	0.565	0.2271	−0.0198, CI [−0.039, −0.003]
`best_single_global` (0.15)	0.155	0.601	0.2469	—
`fitted_per_task`	0.163	0.582	**0.2544**	+0.0075, CI [−0.011, +0.027]

Nine fitted thresholds beat one global threshold by **+0.0075 with a CI straddling zero** — *in sample*, where the nine have nine free parameters to its one. Yet the single global value is *solid*: re-running the sweep inside every bootstrap draw, **0.15 is re-selected in 94%** of them against a 10% chance rate.

Pooled over 135 videos the operating point is identified. Split nine ways over 15 videos each, it is not. **That is a power problem, not a claim that the nine tasks are alike.** Note also that the *shipped* per-task thresholds are significantly *worse* than a flat 0.15 (−0.0354), and the damage is in recall (0.356 vs 0.601): they were strangling the system.

5.4 Finding 4 — pass 2 confirmed a prediction made before the data existed

Before pass 2 finished, a prediction was registered in writing: the refined grid's CIs would contain zero on *at least seven of nine* tasks.

The outcome was 9 of 9. Across each refined neighbourhood the entire F1 span is 0.021–0.133, and on four tasks *every* cell sits within the 0.03 noise band of the best. Refining did not resolve a winner; it confirmed there was no gradient to resolve.

Two honesty notes carried in the record: two rows report CI [+0.000, +0.200] whose lower bound is *exactly* zero and therefore does not exclude it (those tasks match 0–2 events out of 15–16); and pass 2's re-selection of 38–74% looks better than pass 1's 25–61% only until you divide by chance — 20% here against 8–10% there. **As a multiple of chance, pass 2 is no better.**

5.5 Finding 5 — the fit does not overfit

Stage 3's 2,700 samples *contain* the 135 fitting videos, so the headline is ≈5% in-sample. Rescoring with those ids removed — free, the same predictions filtered — gives an in-sample bias of **+0.0007 gross**, two orders of magnitude inside the noise band. Whatever else is true of this fit, it generalises.

6 Stage 3: the full benchmark

At 94% banked (2,553 of 2,700), judge offline:

stratum	n	n_{gt}	n_{emit}	t-P / t-R	**time-F1**
audio required	1643	4916	16185	0.155 / 0.510	0.2375
audio not required	868	3511	12866	0.164 / 0.601	0.2578
gross	2511	8427	29051	0.159 / 0.548	**0.2464**

The system emits **3.4× more often than there are events**. Precision, not recall, is what limits the score. And because precision is flat across the whole grid, the gate's absolute ceiling is $2p/(p+1)$ = **0.274**: every remaining threshold adjustment in existence is worth ≤0.028, i.e. inside the noise band.

6.1 Against the vision-only parent

The parent system (Qwen3-VL-8B, vision only, its own fitted thresholds, and a *working* judge) versus ours, on time-F1 only:

	emits per event	micro time-F1	macro time-F1
system_3 (Qwen3-VL-8B, V), n=931	1.48	**0.2684**	**0.2565**
omni (Qwen2.5-Omni-7B, A+V), n=2511	3.45	**0.2464**	**0.1927**

We win on 2 of 9 tasks. Micro-averaged we trail by 0.022 (inside the noise band); **macro-averaged by 0.064**, which is not. The micro figure flatters us because it is dominated by the tasks we do adequately. The parent is conservative and accurate; we are talkative and vague. *Caveat:* different subsets, different backbone, different modality — and the parent's n is as low as 12 on a task where it beats us most. This is a signpost, not a controlled comparison.

7 Limitations — read this section before quoting any number

1. **Only one of the gate's three knobs was refitted.** `mode` and `refractory_s` are inherited unchanged from the vision-only model. The document's own evidence says they are coupled to the threshold, so this bounds what any threshold result can mean. It shows up concretely: the 600 s / 300 s refractories permit ≈1 emission per video on three tasks, and those are our three worst.

2. **There is no out-of-sample nine-vs-one comparison.** A control arm (flat global 0.15 over all 2,700) was built, preflighted, and then **cancelled before launch** on 1 September 2026. The only fitted-vs-global number is the in-sample +0.0075 above, and it must always be cited as *in sample,* n=135*, nine free parameters against one.* Two predictions registered for that arm are **withdrawn, not resolved**.

3. **15 videos per cell is underpowered**, and the study says so rather than working around it. That is the direct cause of Findings 2–4.

4. **`content_acc` and `joint_f1` are withheld** throughout — the judge is unreachable. The true joint-F1 lies between the lower bound 0.038 and the timing ceiling 0.246, and nothing narrows that without a judge. No comparison to published baselines (MiniCPM-o 4.5 at 0.209 *joint*-F1) is available, and they were judged by a different model on a different subset anyway.

5. **The eval is not bit-reproducible.** Treat <0.03 as noise, always.

6. **And the deepest one:** a separate diagnosis found the controller's ICL schema *collapsing* on this backbone — a mean of 21.7 generated tokens against a 300-token budget, with the self-pacing fields appearing in 20 ticks out of 78,828. The boolean this study thresholds is produced by that generation step. So Finding 1's AUC ≈ 0.5 is very plausibly **an artefact of a broken upstream stage rather than a property of the task**, which would mean this study measured the gate of a system that was not perceiving properly. That does not invalidate the findings — they are correct *about the system as it ran* — but it changes what they predict about a repaired system.

8 What it all means

The honest summary. Per-task threshold fitting did not work, and the study is strong evidence about *why*: not because the tasks are alike, but because the signal being thresholded carries no timing information, and because 15 videos per task cannot identify nine parameters. A single global threshold (0.15) is the only operating point this data identifies, and it is identified well. The shipped per-task thresholds are measurably worse than that single value, so replacing them is a real improvement even though the per-task *fit* is not.

What is worth carrying forward:

- **A negative result, properly established** — by four independent code paths (ROC, flat precision curve, paired bootstrap, refined grid), with predictions registered before the data and *withdrawals recorded as visibly as hits.*

- **The fit generalises** (+0.0007 in-sample bias), so the negative result is about identifiability, not overfitting.

- **The next move is not a better gate.** With precision flat, the gate has ≤0.028 left in it. The gains are upstream: make the controller finish its schema, and use the onset time the model already names on 41% of ticks and the gate currently throws away.

Where the evidence lives. Every number above is recomputed from banked predictions and can be re-derived without a GPU: `PERCEPTION_AUDIT.json` (Finding 1), `FIT_NOISE_AUDIT.json` and `FIT_NOISE_AUDIT_P2.json` (2 and 4), `ABLATION.json` (3), `FINAL_THRESHOLDS.json` (the fitted values), `STAGE3_CONFIG_APPLIED.diff` (what stage 3 actually loaded), `FIRST_TICK_AUDIT.json` and `REFRACTORY_AUDIT.json` (§2). The full running record, including every deviation from the original spec and why, is `THRESHOLD_FIT_RUNBOOK.md` §2.1–2.10.